

Microplastics: The Invisible Threat

Inside Your Body, Inside Our Future

1,000+	5g	Confirmed	2050
Particles ingested weekly	Plastic eaten per week	Found in human brain	More plastic than fish

What Are Microplastics?

Microplastics are plastic particles smaller than 5 millimetres — about the size of a sesame seed. They originate from the breakdown of larger plastic items, synthetic textiles, tires, and personal care products. These particles are now found everywhere: in oceans, soil, air, and even inside human bodies.

The Shocking Discovery

In 2024, researchers at the University of New Mexico found microplastics embedded in human brain tissue. The study, published in Nature Medicine, detected polyethylene and polypropylene particles in brain samples from autopsies. The concentration was higher in brains than in other organs, raising serious concerns about neurological effects.

Where They're Found in Your Body

Microplastics have been detected in human blood (Dutch study, 2022), lungs (Brazilian study, 2021), placentas (Italian study, 2021), and now brains (US study, 2024). A typical person ingests approximately 1,000+ microplastic particles per week through food, water, and air — equivalent to eating a credit card's worth of plastic every 7 days.

Health Risks

While research is ongoing, studies link microplastic exposure to: inflammation, cellular damage, disruption of endocrine function, potential neurotoxicity, and carrying of harmful chemicals (PCBs, heavy metals) into tissues. The full long-term health impact remains unknown, which is itself a cause for concern.

Sources in Daily Life

Primary sources include: plastic water bottles (240,000 particles per litre), plastic tea bags (11.6 billion particles per cup), sea salt (microplastic contamination), seafood (fish ingest plastic), synthetic clothing fibers (700,000 fibers per wash), and household dust (inhaling plastic particles).

What You Can Do

1) Use refillable glass or stainless steel bottles instead of single-use PET. 2) Avoid plastic tea bags — use loose leaf with a metal infuser. 3) Support local recycling programs like Syscycl that prevent plastic from entering the environment. 4) Choose natural fiber clothing over synthetic polyester. 5) Reduce single-use plastic in all forms.

Sources

Nature Medicine 2024 (University of New Mexico), Environment International 2022 (Vrije Universiteit Amsterdam), Science of the Total Environment 2021, WHO Guidelines for Drinking Water Quality, Ellen MacArthur Foundation — The New Plastics Economy.

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